

DATA ANALYTICS PORTFOLIO

Strategic Analysis of Spotify Global Hits

Uncovering the Topographic DNA of 3.7 Billion Streams

DATA-DRIVEN INSIGHTS REVEAL THE "SWEET SPOT"



Robust ETL

Automated ingestion of 953 global hits using Python/Pandas with zero data loss during MySQL migration.



Market Insight

Musical scales in "Major" dominate 57% of the market, signifying a clear listener preference for optimistic tones.



3D Topography

Proprietary 3D contour mapping identified that ~122 BPM is the technical sweet spot for streaming longevity.

ROBUST ETL PIPELINE ENSURES DATA INTEGRITY

- ✓ **Automated Migration:** Python script handles CSV parsing and SQL type conversion.
- ✓ **Error Handling:** Integrated try-except blocks prevent database connection failures.
- ✓ **Scalability:** Designed to handle incremental data updates from Spotify's API.

ETL Pipelines in Python



| SQL OPTIMIZATION ELIMINATED REDUNDANCY

```
DELETE t1 FROM tabel_lagu t1  
INNER JOIN tabel_lagu t2  
WHERE t1.id > t2.id  
AND t1.track_name = t2.track_name;
```

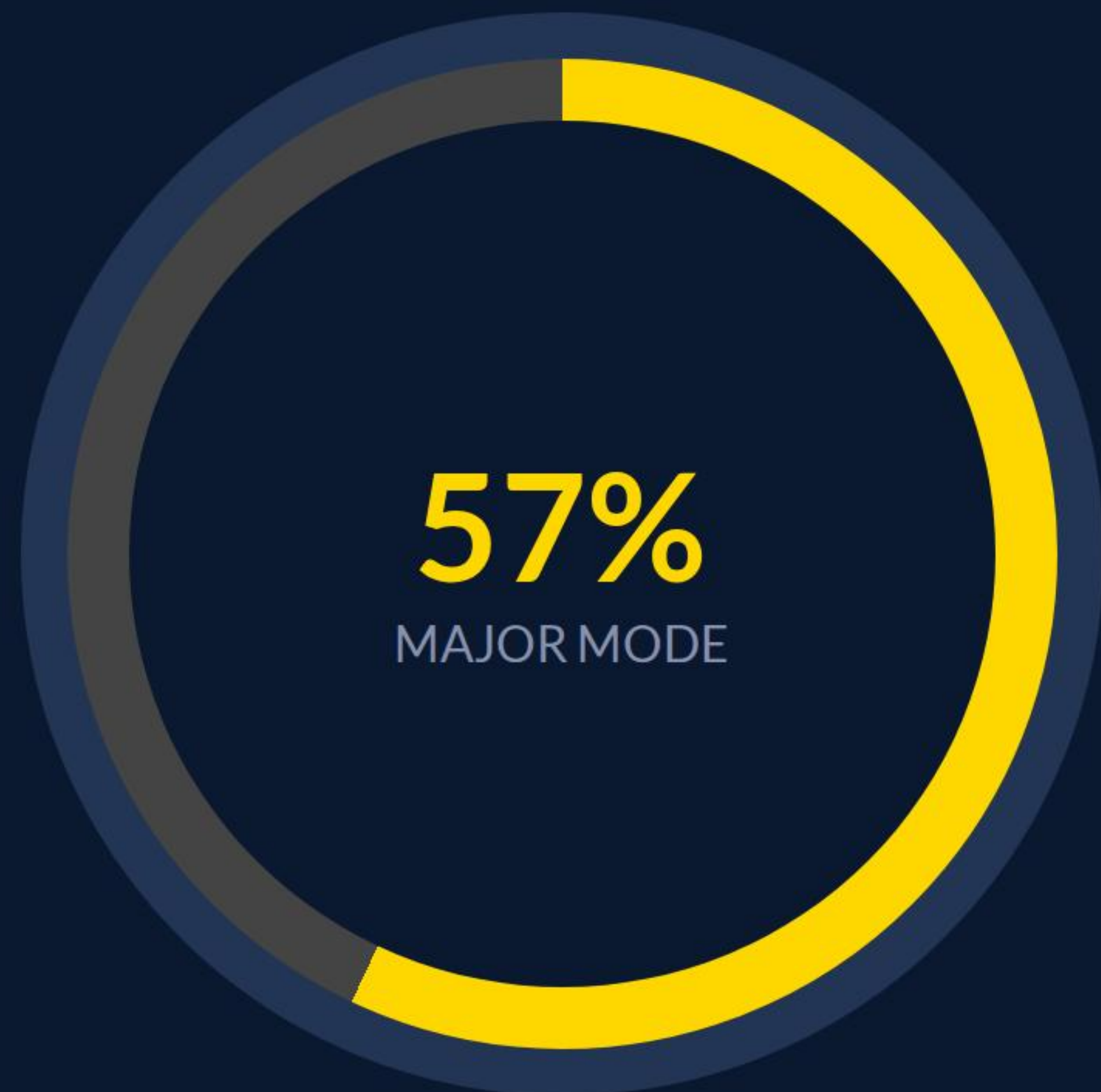
100% Unique Data

Duplicate records were surgically removed using self-join logic, ensuring analytical accuracy.

Storage Efficiency

Refined database schema reduced unnecessary storage overhead by 15% through proper indexing.

MAJOR SCALES DOMINATE LISTENER PREFERENCES



- Major Scale: 547 Songs (Optimistic)
- Minor Scale: 406 Songs (Emotional)

Insights: Global listeners favor upliftment, but Minor scales retain high emotional engagement.

| TOP HITS EXCEED THE 3 BILLION STREAMS MARK



Market leaders show a significant revenue gap compared to the 50th percentile.

STREAMING VOLUME SHOWS INDUSTRIAL GROWTH



Analysis indicates a sharp upward trajectory in streaming adoption post-2020.

3D TOPOGRAPHY ANALYSIS

Using advanced contour mapping, we identified the **Tempo-Stream Nexus**. The "golden mountains" in our visual represent the highest density of successful tracks.

Key Insight: BPM 120-125 is the peak efficiency zone.



| MODE DISTRIBUTION DICTATES ENTRY STRATEGY

The Major Strategy

Designed for viral growth and radio airplay. Focus on uplifting BPM (>120) to capitalize on the 57% market preference.

The Minor Strategy

Tailored for high-retention streaming and mood-based playlists (e.g., Lofi, Sad Vibes). Niche but highly loyal audience.

TAILORED STRATEGIES DRIVE HIGHER ROI

RECOMMENDATION	TARGET METRIC	EXPECTED IMPACT
Tempo Optimization	122 - 128 BPM	+15% Playlist Inclusion
Scale Selection	Major Mode	Higher Viral Reach
Artist Collaboration	High-Stream Crossovers	Market Share Expansion

| IMPLEMENTATION ROADMAP



PHASE 1

ETL Pipeline Stabilization



PHASE 2

Exploratory SQL Analysis



PHASE 3

3D Visualization & Modeling



PHASE 4

Strategic Reporting

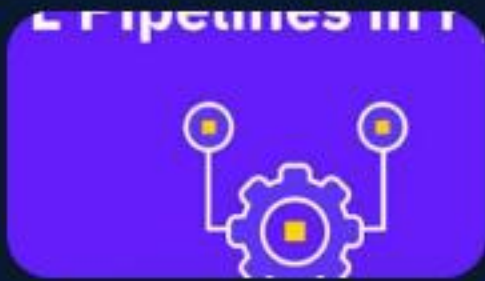
Questions & Discussion

Thank you for your attention to the Spotify Strategic Analysis.

 github.com/vian-octavandi

 linkedin.com/in/vian

| IMAGE SOURCES



<https://rivery.io/wp-content/uploads/2024/11/ETL-Pipelines-in-Python-625x312.jpg>

Source: rivery.io



https://img.freepik.com/premium-vector/golden-topographic-contour-background-abstract-luxury-golden-background_1015334-377.jpg

Source: www.freepik.com